

Natural Systems

Natural systems are systems that exist in nature and are not made by humans. These systems work on their own and follow natural rules.

Examples:

- Weather system • Solar system
- Ecosystem (plants, animals, environment) • Human body

Physical Systems

Physical systems are systems made up of real, visible objects that you can touch and see. These systems involve physical parts working together and are controlled by the laws of physics.

Examples:

- A computer (works using electricity and follows electrical laws).
- A car (moves based on laws of motion and force).
- A washing machine (rotates using mechanical and electrical laws).
- A fan (works due to electricity and motion laws).

Chemical Systems

Chemical systems involve substances that change or react to form new substances.

Examples:

- Water (H_2O) forms from hydrogen and oxygen.
- Acid + base $\rightarrow$ salt + water. This is a chemical reaction

Biological Systems

Biological systems are systems related to living things. They include parts of living organisms that work together to keep them alive and functioning.

Examples:

- Human body with organs like heart, lungs, and brain.
- Circulatory system – transports blood, oxygen, and nutrients.
- Nervous system – controls body activities and responses.
- Immune system – defends against diseases.

Psychological Systems

Psychological systems are systems related to the mind and behavior of humans. They explain how people think, feel, learn, and react to different situations.

Examples:

- Thinking and decision-making.
- Emotions like happiness, fear, and anger.
- Human behavior and reactions.

Artificial Systems

Created by humans to solve problems, perform tasks, or make life easier.

Examples:

- Computers
- Mobile phones
- Cars
- Robots

Knowledge Systems

Knowledge systems are organized ways of collecting, storing, managing, and using information so people can learn new things, solve problems, and make better decisions in daily life.

Example:

- Libraries  (store books and knowledge)

- Schools 🏫 (teach and share knowledge)
- Computers and databases 💻 (store and manage large data)
- Online learning systems (like Google and AI tools)

Types:

- **Mathematics** ➗ :Mathematics is a system that helps us solve problems using numbers, formulas, and calculations.

Examples:

- Finding total bill in shopping
- Calculating exam percentage
- **Logic** 🧠 :Logic is a system that helps us think in a correct and step-by-step way to make good decisions. It helps in reasoning and problem solving

Examples:

- If it is raining → take an umbrella
- If marks are above 50 → student passes
- **Databases** 🗄️ :Databases are systems used to store, organize, and manage large amounts of data in a structured way so it can be easily accessed and updated.

👉 Common database systems include:

- > MySQL
- > NoSQL
- > MongoDB

Examples:

- School student records
- School student records
- Online shopping data
- **Information Management Systems:** An Information Management System is a system that collects, processes, stores, and distributes information in an organized way to help organizations work efficiently.

Examples:

- School management system (students, teachers, results)
- Hospital system (patient records)
- Office data management

Engineering Systems

Engineering systems are man-made systems that use science and technology to solve real-world problems.

Examples:

- Bridges and buildings
- Electrical power systems
- Water supply systems
- Roads and traffic systems
- Machines like elevators or generators
- **Civil Engineering:** These systems are related to buildings and infrastructure. They help in designing and constructing structures for public use.

Examples:

- Roads
- Bridges
- Buildings
- Dams
- **Mechanical Engineering:** These systems involve machines and mechanical devices that move or work using force and energy.

Examples:

- Cars
- Engines
- Fans
- Machines in factories
- **Chemical Engineering:** These systems deal with chemicals and processes that change substances into new materials.

Examples:

- Oil refining
- Fertilizer production
- Medicine manufacturing
-
- **Electrical Engineering:** Deals with electricity and electronic systems, including power grids, home automation, and electric vehicles.

Examples:

- Food processing
- Electrical circuits
- Transformers
- Home wiring systems
- **Software Engineering:** These systems are related to computer programs and software development.

Examples:

- Mobile apps
- Websites
- Operating systems (Windows, Android)
- Computer games

Social Systems

Social systems are systems where people live, interact, and follow rules in a society.

Examples:

- ♦ Family system
- ♦ School system
- ♦ Government system
- ♦ Friend groups
- ♦

- **Academic Institutions:** These are places where people go to learn and gain knowledge.

They are social systems because students, teachers, and staff work together.

Examples:

- Schools
- Colleges
- Universities
- **Governments:** Government is a system that makes rules, manages the country, and provides services to people.

Examples:

- Local government
- Provincial government
- Federal government
- **Organizations:** Include businesses and non-profits. Examples: A corporation like Apple focuses on technology and profit, while an NGO like the Edhi Foundation works for social welfare and helping people in need.